

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Case Study

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	<i>Analyse and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)</i>		
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Section / Batch:	2024	Date:	11-08-2026

Code of Conduct

I _____ Sk. Fawaz Ahamad / 192425389 _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Sk. Fawaz Ahamad _____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyse the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Chennai, Tamil Nadu

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. Tech – Artificial Intelligence & Machine Learning

CO3-AT2

Case Study

Category	Details
Submitted By	Sk. Fawaz Ahamad
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Program	B. Tech Artificial Intelligence & Machine Learning
Year & Semester	III Year
Department	Computer Science and Engineering
Submitted To	Faculty Name: S. Christy
Subject	Computer Vision
Academic Year	2024–2028
Date of Submission	11 August 2026

9. Feature Matching under Illumination Changes

9.1 Introduction

Feature matching is an important technique in Computer Vision used to identify corresponding points or regions between two images of the same scene. It is widely used in applications such as **image stitching, object recognition, image registration, 3D reconstruction, visual tracking, and augmented reality.**

However, when the same scene is captured under different lighting conditions, the appearance of objects may change significantly. For example, an image captured in bright sunlight may contain strong edges and high-intensity regions, while the same scene captured at night may contain weak edges, shadows, and low contrast.

These changes can cause feature detectors and descriptors to produce different feature representations. As a result, correct corresponding features may not be matched, while incorrect features may sometimes be considered as matches.



Case Scenario

Consider two images of the **same building captured during daytime and evening.**

- Image 1: Bright daylight with high illumination.

- Image 2: Low illumination with shadows.
- The scene structure remains the same.
- Pixel intensities, contrast, shadows, and local textures change.

The objective is to understand **why feature matching fails under illumination changes and how preprocessing and robust feature descriptors can improve matching performance.**

9.2 Problem Identification

The major problem is that traditional feature matching methods can be sensitive to changes in image intensity.

When illumination changes:

1. Pixel intensity values change.
2. Bright regions may become dark or saturated.
3. Shadows may appear or disappear.
4. Image contrast may change.
5. Some edges become weaker or stronger.
6. Local textures may look different.
7. Feature descriptors generated from the two images may no longer be sufficiently similar.

Therefore, the matching algorithm may produce:

- **Missing matches**
- **Incorrect matches**
- **Reduced number of correct correspondences**
- **Low matching accuracy**

Example

Suppose a corner has an intensity pattern:

Day:

Bright → Medium → Dark

After illumination changes:

Evening:

Medium → Dark → Very Dark

Although the physical corner is still present, its numerical representation has changed. A simple intensity-based descriptor may therefore consider them different

9.3 Why Feature Matching Fails under Illumination Changes**1. Change in Pixel Intensity**

Lighting directly affects the intensity values of pixels.

For example:

where:

- = original image intensity
- = contrast change
- = brightness change
- = modified intensity

Even though the object has not moved, its pixel values can change substantially.

2. Contrast Variation

A bright image may have strong differences between neighbouring pixels, whereas a dark image may have weak differences.

This can affect:

- Edge detection
- Corner detection
- Texture detection
- Feature descriptor generation

Consequently, the same physical feature may not receive the same descriptor.

3. Shadows and Highlights

Different lighting conditions can introduce new shadows or remove existing ones.

For example:

Original:

Object → uniformly illuminated

Changed

lighting:

Object → partially illuminated + shadow

The shadow can create artificial edges and features, resulting in incorrect matches.

4. Saturation

Very bright illumination can cause some image regions to become saturated.

A saturated region contains little useful intensity information. Therefore, feature detectors may fail to find reliable key points in those areas.

5. Loss of Local Texture

Low illumination can hide fine details and textures.

For example:

Daylight:

Brick patterns, corners and surface textures are clearly visible.

Low

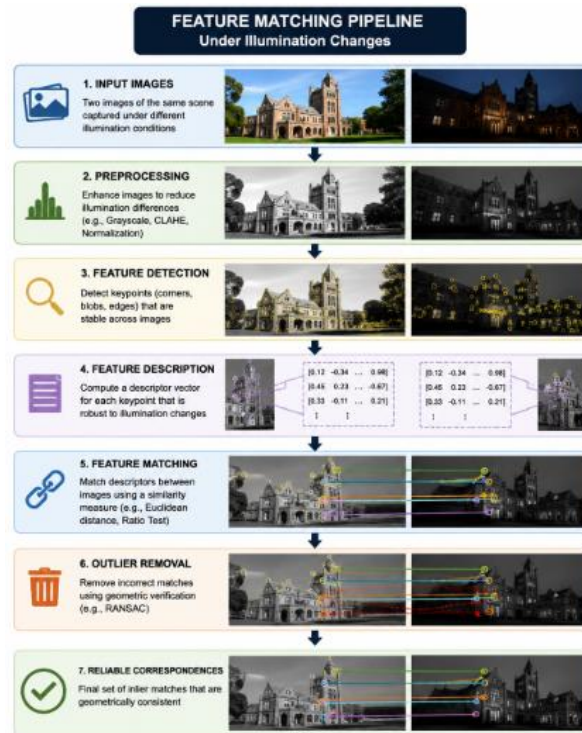
light:

Many of these details become difficult to detect.

This reduces the number of reliable feature points.

9.4 Effect on Feature Detection and Matching

The complete matching process can be represented as:



Under illumination changes, errors can occur at almost every stage.

Stage	Effect of Illumination Change
Preprocessing	Different brightness and contrast
Feature Detection	Key points may be missed
Description	Descriptor values may change
Matching	Correct matches may receive low similarity
Outlier Removal	More incorrect matches may need rejection
Final Result	Reduced matching accuracy

9.5 Preprocessing Techniques

Preprocessing can reduce the effect of illumination differences before feature extraction.

A. Histogram Equalization

Histogram equalization improves the distribution of intensity values and can increase contrast in dark images.

Advantage

It makes important image details more visible and can help feature detectors identify useful key points.

Limitation

It may sometimes over-enhance noise or create unnatural contrast.

B. Contrast Limited Adaptive Histogram Equalization (CLAHE)

CLAHE performs contrast enhancement locally instead of applying the same enhancement to the entire image.

It is particularly useful when an image contains both:

- Bright regions
- Dark regions

It can improve local details while limiting excessive contrast enhancement.

C. Gamma Correction

Gamma correction modifies image brightness using:

It can be used to brighten dark images or reduce excessive brightness.

D. Normalization

Intensity normalization scales pixel values into a common range.

For example:

This reduces differences caused by overall brightness and contrast.

9.6 Robust Feature Descriptors

Preprocessing alone may not completely solve the problem. Therefore, feature descriptors that are more resistant to illumination changes should be used.

1. SIFT

Scale-Invariant Feature Transform (SIFT) detects keypoints and generates descriptors based on local gradient information.

SIFT is relatively robust to:

- Scale changes
- Rotation
- Moderate illumination changes

Instead of depending directly on raw pixel intensity, SIFT uses local gradient information to describe the feature.

Why SIFT helps

If the brightness of an image changes, the exact pixel values may change, but the **local structure and gradient orientation** around an important feature can remain relatively similar.

Therefore, SIFT can produce more reliable matches than simple intensity-based methods.

2. ORB

Oriented FAST and Rotated BRIEF (ORB) combines FAST keypoint detection with a BRIEF-based descriptor.

Advantages include:

- Fast computation
- Low computational cost
- Suitable for real-time applications
- Reasonable robustness to moderate illumination changes

ORB is particularly useful when computational efficiency is important.

3. SURF

Speeded-Up Robust Features (SURF) uses local image structures and gradient information to generate robust descriptors.

It provides good performance under several image transformations, including moderate illumination changes.

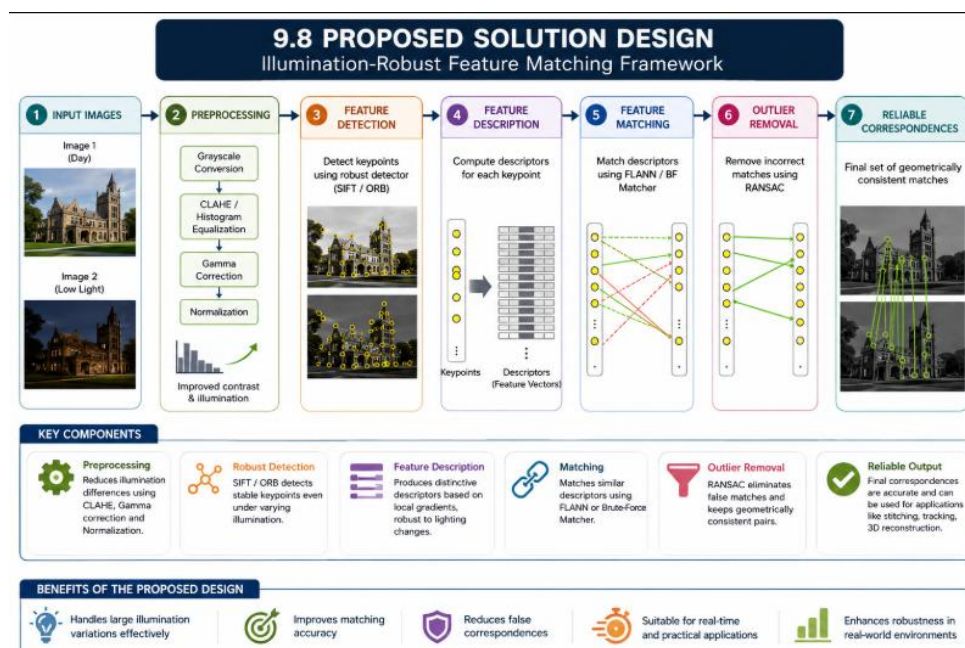
9.7 Comparison of Approaches

Method	Illumination Robustness	Speed	Main Advantage
Raw intensity matching	Low	High	Simple
Histogram Equalization	Moderate	High	Improves contrast
CLAHE	Good	Moderate	Local contrast enhancement
ORB	Moderate–Good	Very High	Fast and efficient
SIFT	High	Moderate	Robust local descriptor
SURF	High	Moderate	Robust and relatively fast

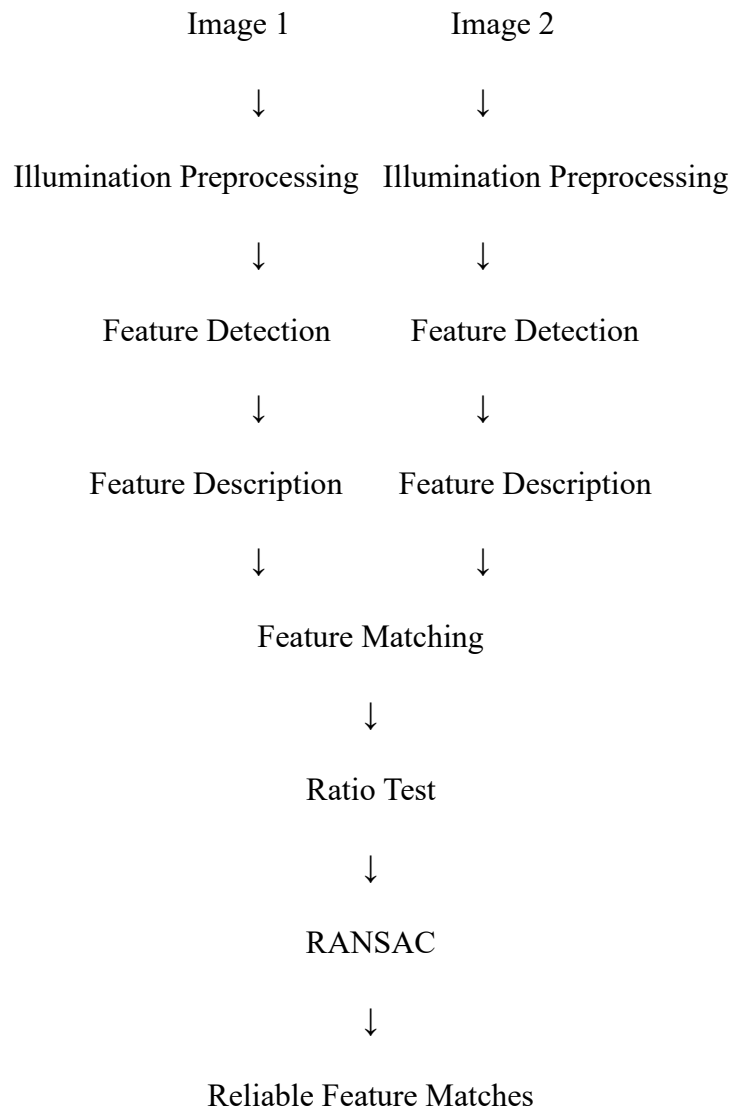
Recommended approach: For a case where illumination differences are significant, combining illumination preprocessing + SIFT/ORB + robust matching + outlier removal provides a stronger solution than relying on raw feature matching alone.

9.8 Proposed Solution Design

A suitable solution for this case is:



9.9 Proposed Workflow



9.10 Expected Improvement

The proposed approach improves matching because preprocessing reduces the visual difference caused by illumination, while robust descriptors capture more stable local image structures.

Without preprocessing / robust descriptor

Day Image → Feature Extraction → Poor Correspondence → Incorrect/Missing Matches

With proposed approach

**Day Image → Normalization/CLAHE → SIFT/ORB → Robust Matching → RANSAC
→ Reliable Matches**

Therefore, the number of correct matches is expected to increase, while incorrect matches are reduced.

9.11 Practical Example

Consider a surveillance camera observing a road during daytime and nighttime.

During daytime:

- Road markings are clearly visible.
- Buildings have strong edges.
- Vehicle features are easy to detect.

During nighttime:

- Overall intensity is reduced.
- Shadows become stronger.
- Some regions become completely dark.
- Headlights may create highly bright regions.

If direct feature matching is used, many correspondences may fail.

By applying **CLAHE or normalization** followed by **SIFT/ORB feature extraction and RANSAC-based outlier removal**, stable features such as corners, edges and local structures can be matched more reliably.

This approach can be useful in:

- Surveillance systems
- Autonomous vehicles
- Image registration
- Panorama generation
- Object recognition
- Robotics
- Augmented reality

9.12 Analysis

The case demonstrates that feature matching is not determined only by the physical similarity of two scenes. The **visual representation of the scene can change significantly due to illumination**.

A basic feature-matching system that depends heavily on intensity values may therefore produce unreliable correspondences.

The analysis shows that the problem can be addressed at two levels:

Preprocessing Level

Reduce illumination differences using:

- Histogram Equalization
- CLAHE
- Gamma Correction
- Intensity Normalization

Feature Level

Use descriptors that rely more on stable local structures and gradients, such as:

- SIFT
- ORB
- SURF

Finally, matching techniques such as the **ratio test and RANSAC** can eliminate ambiguous and geometrically incorrect matches.

Thus, combining these techniques provides a more reliable feature-matching system than using a single technique alone.

9.13 Conclusion

Feature matching under different illumination conditions is challenging because changes in brightness, contrast, shadows, highlights, saturation and local texture can alter the appearance of corresponding features.

The problem can be reduced by applying suitable **image preprocessing techniques** and using **robust feature descriptors**. CLAHE, normalization and gamma correction can reduce illumination differences, while SIFT, ORB and SURF can provide more stable feature representations.

A complete solution using:

Preprocessing → Robust Feature Detection → Descriptor Extraction → Feature Matching → Ratio Test → RANSAC

can significantly improve the reliability of feature correspondences.

Therefore, illumination-aware preprocessing combined with robust feature descriptors is an effective solution for accurate image comparison under changing lighting conditions.